

# How do we Improve Big Models?

apart from blood, sweat and tears

- Algorithmically.
  - For example, use an adjoint model to determine what some aspect of the model is sensitive to (be it numerical resolution, a physical parameterization).
- By tuning, and intuition.
  - May seem ad hoc at times, but much experience goes into it.
- By improving the ‘sub-grid’ parameterizations.
  - Need to understand the process because the whole point is not to use a brute force simulation.
- By understanding models’ overall behaviour — and that requires the use of a hierarchy simpler models (that is, using theory!)
  - Suppose a complex model gives the wrong answer — how should we fix it?
  - Just as biologists have fruit flies, climate modellers must construct their own simpler systems to give better understanding (I. Held).